Continuous Integration

Cohort 2 Team 9
Charlie Armstrong
Adam Carey
Matthew Havers
Matthew Kelsey
Daniel Packer
Niall Potts
Sam Savidge
Ke Wu

To implement continuous integration for our project, an autobuild function was set up using GitHub actions that is triggered after each commit in the master branch of our GitHub repository. We used GitHub actions as our repository was already hosted on GitHub, thereby allowing for easy integration with the autobuild function. This function is split into three different pipelines for the three operating systems our game was required to work on: linux was tested via the Ubuntu distribution, windows and MacOS. Each pipeline takes the master branch as input, and outputs if the autobuild function has built successfully, the executable JAR and Jacoco testing report. The testing report is attached to get an overview of the effectiveness of our tests, and helps us identify if we're missing anything.

A release function is triggered by committing a change to the master branch on GitHub that has a version tag on it, this is also split up into three pipelines depending on the version. It takes the master branch as an input and outputs the executable jar into the releases part of the github page.

These pipelines are implemented in Github actions in a YAML file called Gradle.yml. The autobuild pipelines first gives the read permission to the pipeline and makes the gradlew executable by running "chmod +x ./gradlew", then it sets up jdk 17 to be used with "actions/setup-java@v4" from temurin.

Then it builds the project with a Jacoco test report with "./gradlew build jacocoTestReport", and outputs the Jacoco report as an artifact with "actions/upload-artifact@v6.0.0". Finally it uploads the executable JAR with "actions/upload-artifact@v6.0.0" as well.

This process happens for each operating system chosen by the matrix os: [ubuntu-latest, windows-latest, macos-latest].

The release pipeline first gives the write permission to the pipeline, it then downloads the executable using actions/download-artifact@v7.0.0 and finally uploads it to the releases page "withsoftprops/action-gh-release@v2".